Kantesh Kumar

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ResearchGate Profile: https://www.researchgate.net/profile/Kantesh-Kumar
Google Scholar Profile: https://scholar.google.com/citations?view_op=list_works&hl=en&user=VZbnayIAAAAJ
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EDUCATION

Harvard T.H. Chan School of Public Health | Boston, MA, US | *Master of Public Health* | *Presidential Scholar (full-tuition merit award)* May 2027
Aga Khan University | Karachi, PK | *Bachelor of Medicine; Bachelor of Surgery* Feb 2023
Aga Khan Higher Secondary School | Hyderabad, PK | *Higher Secondary School Certificate* May 2018

LICENSING EXAMINATIONS

USMLE STEP 1: PASS (1st attempt, March 2023)

PUBLICATIONS

- **Kumar K,** Motwani S, Abdul Rahim K, Khan Bozai MM, Hassan S, Ahmed T, Mushtaq S, Atiq H, Shafiq Y, Shamim S, Haider A, Razzak J. Outcomes of Moderate-to-Severe Traumatic Brain Injury in a Low-Resource Setting. JAMA Netw Open. 2026 Aug 3;9(8):e2628403. doi: 10.1001/jamanetworkopen.2026.28403. PMID: 42579276; PMCID: PMC13463165.
- **Kumar K,** Abdul Rahim K, Ahmad M, Ali R, Hassan S, Ahmed T, Mushtaq S, Shamim MS, Babar Z, Shafiq Y, Atiq H, Haider A, Razzak J. Long-Term Mortality and Quality of Life Among Adults With Mild Traumatic Brain Injury. JAMA Netw Open. 2026 Aug 3;9(8):e2629041. doi: 10.1001/jamanetworkopen.2026.29041. PMID: 42593789; PMCID: PMC13474037.
- Ghazi, K. R., Arif, A., **Kumar, K.**, Rahim, K. A., Ahmad, S., Jarrar, Z., Mahmood, S. B. Z., Bakhshi, S. K., Ali, M., Samad, Z., & Haider, A. H. (2026). When the clock strikes eight: Nighttime admissions and worse outcomes of patients admitted with emergency general surgery conditions. *The American Journal of Surgery*, 117266. Advance online publication. https://doi.org/10.1016/j.amjsurg.2026.117266
- **Kumar, K.**, Ghazi, K. R., Rahim, K. A., Musharraf, M. B., Hassan, S., Sheikh, S. A., Shafiq, Y., Atiq, H., Ahmed, T., Mushtaq, S., Haider, A. H., & Razzak, J. (2026). *Long-term outcomes following traumatic injury in older adults in Pakistan: A prospective cohort study*. Journal of Surgical Research. https://doi.org/10.1016/j.jss.2026.02.015
- Rahim, K. A., **Kumar, K.**, & Equi-Injury Pakistan. (2026). *Paying the price: The intersection of out-of-pocket expenses and trauma mortality in Pakistan*. World Journal of Surgery. https://doi.org/10.1002/wjs.70243
- Khan, M., Khan, D. A., **Kumar, K.**, Hasan, S. A. T., Khan, S., Najam, F., Mehmood, M., & Nawaz, W. (2026). Epidemiologic Trends in Secondary Malignant Neoplasms of the Liver in the United States: A 25-Year National Study. *Journal of gastrointestinal cancer*, *57*(1), 5. https://doi.org/10.1007/s12029-025-01391-w
- Rahim, K. A., Zafar Mahmood, S. B., Ghazi, K. R., Arif, A., **Kumar, K.**, Bakhshi, S. K., Ali, M., Samad, Z., & Haider, A. A Matched Comparison of Patient Outcomes in Emergency General Surgery Conditions: Understanding Variability in Operative and Nonoperative Cases. World Journal of Surgery. https://doi.org/10.1002/wjs.12539
- Abdul Rahim, K., Ghazi, K. R., Arif, A., **Kumar, K.**, Mahmood, S. B. Z., Bakhshi, S. K., Ali, M., Samad, Z., & Haider, A. H. (2025). Weekends at risk: Uncovering the impact of the weekend effect on emergency general surgery. *Surgery*, *181*, 109275. Advance online publication. https://doi.org/10.1016/j.surg.2025.109275
- Musharraf, M. B., Abdul Rahim, K., **Kumar, K.,** Sheikh, S. A., Hassan, S., Merchant, A. A. H., Shaukat, N., Atiq, H., & Ahmed, T. (2025, April 22). Bimodal, but not the same: Persistent late peaks in trauma mortality. World Journal of Surgery. https://doi.org/10.1002/wjs.12592
- Abdul Rahim, K., Ghazi, K. R., **Kumar, K.**, Arif, A., Mahmood, S. B. Z., Bakhshi, S. K., Zafar, H., & Haider, A. H. (2025). When Is Old NOT Gold? Identifying Global Age Risk Surgical Thresholds to Improve Outcomes. The Journal of Surgical Research, 312, 177–184. Advance online publication. https://doi.org/10.1016/j.jss.2025.05.015

- Shafiq, Y., Muhammad, A., Noor, S., **Kumar, K.**, Chhipa, U. E. A., Valente, M., Razzak, J., Ragazzoni, L., Barone-Adesi, F., & Pham, P. (2025). The modification of risk assessment ranking when considering the female population and children in the INFORM risk index for selected countries. *Scientific reports*, *15*(1), 26527. https://doi.org/10.1038/s41598-025-10490-4
- Shafiq, Y., Muhammad, A., **Kumar, K.**, Wajid Ali, Z., Noor, S., Suhag, Z. H., Tahir, R., Jan, A., Ragazzoni, L., Barone-Adesi, F., & Valente, M. (2025). Toward Resilient Maternal, Neonatal, and Child Health Care: A Qualitative Study Involving Afghan Refugee Women in Pakistan. Health services insights, 18, 11786329241310733. https://doi.org/10.1177/11786329241310733
- **Kumar, K.**, Hassan, S., Musharraf, M. B., Rahim, K. A., Sheikh, S. A., Atiq, H., Ahmed, M. W., Siddiqui, R. U., Naseer, R., Akhter, S., Shafiq, Y., & Razzak, J. (2025). Epidemiological characteristics of elderly population receiving pre-hospital emergency care after road traffic injuries in Punjab, Pakistan. *International journal of emergency medicine*, *18*(1), 140. https://doi.org/10.1186/s12245-025-00898-0
- Ali, U., Ghazi, K. R., **Kumar, K.**, Kanza, S., & Shaikh, S. A. (2025). From injury to reconstruction, utilizing staged technique in a self-inflicted facial gunshot wound: a case report. *Journal of medical case reports*, *19*(1), 544. https://doi.org/10.1186/s13256-025-05414-9
- Sheikh, S. A., Musharraf, M. B., Rahim, K. A., **Kumar, K.**, Hassan, S., Shafiq, Y., Ahmed, M. W., Shahnaz, D., Naseer, R., Razzak, J. A., & Atiq, H. (2025). Leveraging emergency medical services data to analyse characteristics and risk factors of violence: an epidemiological study. *Injury prevention: journal of the International Society for Child and Adolescent Injury Prevention*, ip-2025-045704. Advance online publication. https://doi.org/10.1136/ip-2025-045704
- Abdul Rahim, K., **Kumar, K.**, Arif, A., Ghazi, K. R., Mahmood, S. B. Z., Bakhshi, S. K., Ali, M., Samad, Z., & Haider, A. (2025). Does Time Matter? Impact of Residency Start Month on EGS Outcome in Pakistan. *World journal of surgery*, 10.1002/wjs.70049. Advance online publication. https://doi.org/10.1002/wjs.70049
- Ahmed, S., Waqar, U., **Kumar, K**., & Khan, D. A. (2021). A Bibliometric Analysis of Global COVID-19 Research. *National Journal of Health Sciences*, *5*(3), 119-126. Retrieved from https://ojs.njhsciences.com/index.php/njhs/article/view/173
- Waqar, U., Ahmed, S., Khan, D. A., **& Kumar, K.** (2020). Case-Fatality Rates for COVID-19: Are We Ignoring Something?. *National Journal of Health Sciences*, *5*(2), 89-90. Retrieved from https://ojs.njhsciences.com/index.php/njhs/article/view/10
- Nai, N., Bobba, A., Oberoi, M., Sohail, A. H., **Kumar, K.**, Mozaffar, A. M., Vummanenni, S., Gangu, K., Khan, M. S., & Sheikh, A. B. (2024). Outcomes of COVID-19 Disease in Comparison with Influenza in Renal Transplant Recipients: Results from a Large Nationwide Research Network in the United States. Transplantation proceedings, 56(1), 87–92. https://doi.org/10.1016/j.transproceed.2023.11.020
- **Kumar, K.**, Sheikh, S. A., Ghazi, K. R., Rahim, K. A., Musharraf, M. B., Hassan, S., Shafiq, Y., Atiq, H., Ahmed, T., Mushtaq, S., Haider, A., & Razzak, J. (Under review). Long-term functional status and quality of life in older adults following traumatic injuries: A longitudinal study in an LMIC. Submitted to the World Journal of Surgery.
- **Kumar, K.**, Hassan, S., Rahim, K. A., Waqas, S., Ahmed, T., Mushtaq, S., Atiq, H., Shafiq, Y., Haider, A., & Razzak, J. (2025). (Under review) Does time of admission matter? A comparative study of trauma care in a lower-middle-income country. Submitted to the Journal of Trauma Surgery and Acute Care Open.
- **Kumar, K.**, Ahmad, M. W., Siddiqui, R. U. R., Musharraf, M. B., Ahmad, M., Rahim, K. A., Hassan, S., Sheikh, S. A., Naseer, R., Akhtar, S., Atiq, H., Shafiq, Y., & Razzak, J. (Under review). RTI hotspot mapping and analysis in Punjab, Pakistan. Submitted to the Journal of Pakistan Medical Association.
- **Kumar, K.**, Wajidali, Z., Atiq, H., Chaand, Z., Hassan, S., Khalid, A. R., Chhipa, U. A., Shah, S., Shafiq, Y., Haider, A., & Razzak, J. (Under review). Unrecognized First Responders: A Qualitative Exploration of Lady Health Workers' Role in Disaster Response. Submitted to the Journal of Global Public Health.
- Ahmad, M., **Kumar, K.**, Rahim, K. A., Wajid Ali, Z., Sheikh, S. A., Hassan, S., Farooqi, W., Zampino, S., Sirney, A., Atiq, H., & Razzak, J. A. (2025). (Under review) From obstetrics to cardiopulmonary resuscitation (CPR): Mobilizing Pakistani Lady Health Workers (LHWs) for cardiac emergency response through CPR integration. Submitted to BMJ Open.
- **Kumar, K.**, Wajidali, Z., Hiranand, N., Abdul Rahim, K., Razzak Ghazi, K., Musharraf, M. B., Khan Bozai, M. M., Babar, Z., Abbasi, A., Hassan, S., Bakshi, S., Shamim, M. S., Haider, A., & Razzak, J. (2026). Gaps in the chain of survival for traumatic brain injury care in a resource-constrained health system. Submitted to The Lancet

Regional Health – Eastern Mediterranean.

Book Chapter

- **Kumar, K.**, Rahim, K. A., & Razzak, J. (in press). Injury. In H. Frumkin (Ed.), *Environmental health: From global to local* (4th ed.). Wiley.

ABSTRACTS, PRESENTATIONS, AND POSTERS

- Hassan, S., **Kumar, K.**, Ahmad, M., Rahim, K. A., Babar, Z., Wajidali, Z., Atiq, H., Haider, A. H., & Razzak, J. A. (2026). Interpersonal violence trauma in Pakistan: Epidemiology, severity, and acute care needs across a multicenter cohort [Abstract 605]. *Annals of Emergency Medicine*, *88*(4), S272–S273. https://doi.org/10.1016/j.annemergmed.2026.08.621
    - Abstract Presented at the American College of Emergency Physicians Research Forum 2026
- **Kumar, K.**, Hassan, S., Rippon, B., & Razzak, J. A. (2026). Geographical differences in burden of emergency medical disease in the United States [Abstract 407]. *Annals of Emergency Medicine*, *88*(4), S189. https://doi.org/10.1016/j.annemergmed.2026.08.419
    - Abstract Presented at the American College of Emergency Physicians Research Forum 2026
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2026
- **Kumar, K.**, Wajidali, Z., Hiranand, N., Rahim, K. A., Ghazi, K. R., Chaand, Z. B., Hassan, S., Haider, A. H., & Razzak, J. A. (2026). Gaps in the chain of survival for traumatic brain injury care in a resource-constrained health system.
    - Abstract Quick Shot Presentation at American College of Surgeons Clinical Congress 2026
- Ghazi, K. R., Ahmad, S., Khowaja, A. H., Rahim, K. A., **Kumar, K.**, Nasim, M. T., & Haider, A. H. (2026). Who fails to be rescued? Predictors of mortality after complications in emergency general surgery in Pakistan.
    - Abstract Quick Shot Presentation at American College of Surgeons Clinical Congress 2026
- Hassan, S., **Kumar, K.**, Rahim, K. A., Wajid Ali, Z., Chand, Z., Atiq, H., Belli, A., & Razzak, J. A. (2026). Disparities in traumatic brain injury outcomes: A comparative study of urban and rural Pakistan.
    - Abstract Presented at the Society for Academic Emergency Medicine Annual Meeting 2026, Atlanta, GA (Abstract 736)
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2026
- Garaffa, J., **Kumar, K.**, Ansari, M., Ashfaq, L., Hassan, S., Akhter, S., Naseer, R., Khan, M., Atiq, H., & Razzak, J. A. (2026). Epidemiological analysis of probable suicides reported by Emergency Medical Services in Punjab, Pakistan.
    - Abstract Presented at the Society for Academic Emergency Medicine Annual Meeting 2026, Atlanta, GA (Abstract 584)
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2026
- Hassan, S., Motwani, S., **Kumar, K.**, Rizwan, H., Rizvi, N., Tasleem, A., Manan, A., Amanullah, S., Razzak, J. A., & Shafiq, Y. (2026). Who calls for help? Lower female utilization of prehospital Emergency Medical Services in Punjab, Pakistan.
    - Abstract Presented at the Society for Academic Emergency Medicine Annual Meeting 2026, Atlanta, GA (Abstract 453)
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2026

- Kumar, K., Chand, Z., & Razzak, J. A. (2026). Governance and capacity gaps in heat action plan implementation: Insight from a high-risk, low-resource setting.
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2026
- Rahim, A. K., Ghazi, R. K., **Kumar, K.,** Arif, A., Mahmood, Z. S., Bakhshi, S., Ali, M., Sarmad, Z., Haider, A. When is Old NOT Gold? Identifying Global Age Risk Surgical Thresholds to Improve Outcomes.
    - Abstract Quick shot Presentation at 20th Annual Academic Surgical Congress 2025
- Sheikh, S. A., Musharraf, M. B., Rahim, K. A., **Kumar, K.,** Atiq, H., Shafiq, Y., Murtaza, N., Shahnaz, D., Naseer, R., & Razzak, J. Leveraging Emergency Medical Services data to analyze characteristics and risk factors of violence: An epidemiological study
    - Abstract Quick shot Presentation at 20th Annual Academic Surgical Congress 2025
- **Kumar, K.,** Ghazi, R. K., Rahim, A. K., Musharraf, B. M., Hassan, S., Akhtar, S., Shafiq, Y., Atiq, H., Razzak, J.A (2024). Long-term Mortality after Older Adults' Trauma in Low-Middle Income Countries
    - Abstract Orally Presented at 20th Annual Academic Surgical Congress 2025
- Hassan, S., **Kumar, K.**, Musharraf, B., Abdul Rahim, K., Akhter, S., Shafiq, Y., Atiq, H., & Abdul Razzak, J. Rapid response in prehospital care: Unveiling patterns and perils in traumatic brain injuries in Pakistan.
    - Abstract Orally Presented at Society for Academic Emergency Medicine 2025
- Abdul Rahim, K., **Kumar, K.**, Musharraf, B., Hassan, S., Huba, H., Zabin, Z., Zaheer, Z., Central Team, Haider, A., & Abdul Razzak, J. Paying the price: The intersection of out-of-pocket expenses, mortality, and healthcare inequities.
    - Abstract Orally Presented at Society for Academic Emergency Medicine 2025
- **Kumar, K.**, Hassan, S., Abdul Raheem, K., Waqas, S., Shafiq, Y., Atiq, H., & Razzak, J. The circadian effect in trauma care: A comparative analysis based on time of admission in a lower-middle-income country.
    - Abstract Oral Presentation at Society for Academic Emergency Medicine 2025
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2025
- **Kumar, K.**, Sheikh, S. A., Ghazi, K. R., Abdul Rahim, K., Musharraf, M. B., Hassan, S., Shafiq, Y., Atiq, H., Haider, A., & Razzak, J. Long-term functional status and quality of life in older adults following traumatic injuries: A longitudinal study in an LMIC.
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2025
- **Kumar, K.**, Abdul Rahim, K., Ahmad, M., Ali, R., Hassan, S., Shafiq, Y., Atiq, H., Haider, A., & Razzak, J. Short-term and long-term outcomes of mild TBI patients in LMICs.
    - Abstract Poster Presentation at Weill Cornell Emergency Medicine Research Symposium 2025
- Hiranand, N., **Kumar, K.**, Ibrahim, M. T., Khaskheli, S. A., Majeed, H., Arshad, A. (2024). Sleepless hospital nights: Patients' sleep at a tertiary care hospital
    - Poster Presented at Students' Perspectives In Research & Innovation Trends SPIRIT Conference 2024
- Babar, Z., **Kumar, K.**, Razzak, J. (2025). Building Heat Resilience in Pakistan's Most Vulnerable Districts: A Local Government-Led Approach
    - Presented at the International Humanitarian Studies Association Conference 2025 in Istanbul, Turkey
- Ahmad, M., **Kumar, K.**, Hiranand, N., Soomro, A. Y., Razzak, J. Insights from Lady Health Workers and Communities on Disaster Response in Shaheed Benazirabad District, Pakistan.
- Podium talk given at American Muslim Medical Student Association National Conference (2025)

- Ahmad, M., **Kumar, K.,** Rahim, A.K., Wajid Ali, Z., Sheikh A. S., Farooqi W., Hassaan, S., Sirney A., Atiq, H., Razzak, J.A. (2024). From Obstetrics to CPR: Mobilizing Pakistani Lady Health Workers for Cardiopulmonary Resuscitation.
    - Abstract Orally Presented at Weill Cornell Medicine Emergency Medicine Symposium 2024
    - Poster Presented at Weill Cornell Medicine International Medical Student Education Global Health Reception
    - Poster Presented at New York Academy of Medicine's 2024 Advocacy in Medicine Conference, Navigating Borders: Addressing Global and Immigrant Healthcare Needs
    - Poster Presented at Weill Cornell Medicine 2024 Medical Student Research Day
    - Abstract Orally Presented at APPNA Winter Meeting 2024
    - Abstract Orally Presented at International Conference on Emergency Medicine 2025
- **Kumar, K.,** Shah, S., Hassan, S., Chhipa, U., Wajidali, Z., Atiq, H., Shafiq, Y., Haider, A., Razzak, J.A. (2024). Beyond the Call of Duty: Experiences and Role of Lady Healthcare Workers in Disaster Response.
    - Poster Presented at Weill Cornell Medicine Emergency Medicine Symposium 2024
    - Abstract Orally Presented at Society for Academic Emergency Medicine Conference 2024 – Author in presentation.
- LHW Training in Disaster Response in Rural Sindh – Speaker – Emergency Medicine Education & Research Global Exchange EMERGE 2024 Conference
- **Kumar, K.,** Hassan, S., Musharraf, B. M., Rahim, A. K., Atiq, H., Akhtar, S., Razzak, J.A. (2024). Epidemiological Characteristics of the Elderly Population Involved in Road Traffic Injuries in Punjab, Pakistan
    - Poster Presented at Weill Cornell Emergency Medicine Symposium 2024
- Musharraf, B. M., Rahim, A. K., **Kumar, K.,** Sheikh, A. S., Hassan, S., Merchant, A., Shaukat, N., Atiq, H., Haider, Adil., Razzak, J. A. (2024). In-Hospital Mortality Patterns in Trauma: A Retrospective Analysis from South Asia.
    - Poster Presented at Weill Cornell Medicine Emergency Medicine Symposium 2024

RESEARCH EXPERIENCE

Centre of Excellence for Trauma & Emergencies | Karachi, PK | *DCRF Research Fellow* Mar 2024 – March 2025
Centre of Excellence for Trauma & Emergencies | Karachi, PK | *Research Associate* Mar 2025 – July 2026

- Execute diverse global health projects at Aga Khan University to explore outcomes of Pakistani trauma patients and Emergency Medical Services
- Conducted and presented literature review findings to support the research team in contextualizing data and analysis
- Developed Disaster Preparedness and Response Leadership Course – 1-year course focusing on disaster management techniques for managers and leaders – Participants include private and public sector employees directly or indirectly involved in Disaster Management.
- Acted as the Coordinator for the Disaster Preparedness and Response Leadership Course Jan 2025-Jan 2026. Currently planning
- Utilize R, STATA, and NVivo to analyze quantitative and qualitative datasets across several projects
- Mentored several Weill Cornell Medical Students in Project Design, ERC submission, Data collection, Analysis, and Manuscript Writing

Traumatic Brain Injury Research Program

- Led analysis for two first-author cohort studies published in JAMA Network Open (Aug 2026), quantifying 12-month mortality and quality-of-life outcomes among trauma registry patients with moderate-to-severe and mild

TBI in Karachi

- First-authored a qualitative study mapping barriers to TBI care using the Four Delay model, incorporating perspectives from neurosurgeons, EM physicians, and EMS providers to identify candidate interventions (submitted, Lancet Regional Health – Eastern Mediterranean)
- Currently leading a feasibility study: used the Delphi process to narrow candidate interventions to 3, now assessing feasibility across 4 hospitals in Sindh via structured checklists and in-depth interviews

EMS Data & Trauma Epidemiology

- Clean, manage, and analyze national EMS data (Rescue 1122, Punjab) to answer targeted trauma epidemiology questions
- First authored a study on epidemiological characteristics of elderly patients receiving pre-hospital care after road traffic injuries in Punjab (International Journal of Emergency Medicine, 2025)

Climate & Health

- Contributed to Elrha-funded HEAT trial dissemination: produced a public-facing video translating trial findings into community health messaging; organized a Heat Consultative Meeting to align district stakeholders on Heat Action Plan (HAP) development
- Developed the foundational HAP framework and worked directly with Sindh's five most heat-vulnerable districts to build individualized HAPs; organized a follow-up consultative meeting presenting completed HAPs as a feasibility model for the remaining districts, informing provincial (PDMA) documentation
- Leading a qualitative follow-on study on stakeholder-reported barriers to HAP implementation (data collection complete; analysis pending)
- Leading a separate study modeling the association between EMS activations and ambient temperature using Distributed Lagged Non-linear Models (DLNM); analysis complete, manuscript in preparation

Disaster Preparedness & Response

- Co-developed a three-tier training program — 1-day and 4-day trainings for Lady Health Workers/Healthcare Workers, and a 1-year Disaster Preparedness and Response Leadership Course for managers and sector leaders; served as Course Coordinator (Jan 2025 – Jan 2026)
- Designed and ran simulation drills to give leadership course participants hands-on disaster-response experience
- Led a district-level disaster drill exercise, engaging multiple stakeholder roles to stress-test emergency response systems and surface gaps
- Evaluating LHW/HCW training effectiveness using the Kirkpatrick Model — Reaction and Knowledge via pre/post, knowledge retention re-tested at 6–12 months, and Behavior change via post-training focus group discussions

Mentorship

- Mentored Weill Cornell Medicine students during their research visits to Pakistan across project design, ERC/IRB submission, data collection, analysis, and manuscript writing.

Grants

- Composed and submitted NIH R01 grant focusing on improving Traumatic Brain Injury (TBI) outcomes in low-resource settings under PI mentorship
- Composed and submitted a proposal for intramural funding for developing an MNCH Emergency Vulnerability Index by utilizing Emergency Medical Services data and Demographic Health Survey Data.
- Composed and submitted a Wellcome Trust Grant for Climate Impact Awards.

CLINICAL EXPERIENCE

Aga Khan University Hospital | Plastic Surgery | Selective Jul 2021 – Aug 2021

- Obtained and documented patient history and examinations to brainstorm differential diagnoses with the medical team
- Assisted in melanoma and sebaceous cyst removal, abdominoplasty, and facial nerve transplantation procedures

Aga Khan University Hospital | Cardiothoracic Surgery | Elective Sep 2021 – Oct 2021

- Participated in rounding and procedures (coronary artery bypass grafting, thoracotomies, and tumor resections)

- Observed utilization of medical technologies such as ECG, Echo, and Myocardial perfusion scan in the Cath lab

University of New Mexico | Trauma Surgery | Sub-Internship Aug 2023 – Sep 2023

- Received patients in the ER and performed primary and secondary patient surveys to inform the medical care plan
- Participated in procedures, including exploratory laparotomies, internal rib plating, and split-skin grafts
- Assisted with wound dressing and staple removal to provide prompt bedside care to patients

University of Minnesota | Plastic Surgery | Observership Sep 2023 – Oct 2023

- Observed major plastic procedures, including breast reduction, breast implantation, breast reconstruction, and chest wall reconstruction
- Documented operative notes, morning rounds, and outpatient clinic visits to meticulously track patient status

Memorial Hermann Katy Hospital | General Surgery | Elective Oct 2023 – Nov 2023

- Observed OR procedures, including cholecystectomies and hernia repairs, alongside medical and nursing staff
- Assisted in minor inpatient procedures and provided administrative support by documenting patient notes

EXTRACURRICULAR ACTIVITIES AND LEADERSHIP

Aga Khan University | Karachi, PK | *Campus Photographer* Jan 2019 – Jun 2021

- Served the university campus-wide by photographing events for students and the institution as a whole
- Prepared images for publication by editing using Adobe Photoshop and updating institutional websites

Surgery Interest Group, Aga Khan University (AKU) | Karachi, PK | *General Body Member* 2020 – 2021

- Hosted events featuring prominent surgeons from AKU and beyond to network with students
- Served as a volunteer at AKU's annual surgical conference, focusing on the scope of surgery in trauma care

Aga Khan University Sports Olympiad | Karachi, PK |
Media Director (2021)

- Plan and execute media campaigns for different platforms
- Manage the advertising department in different projects undertaken by the sports society
- Use analytics to improve outreach plans
- Understand and manage expectations and liaise with university management to ensure fulfillment of set goals

Media Team Member (2020)

- Provided photography coverage during sports Olympiad events to document institutional memories
- Created video content by recording and editing team member interviews speaking about their experiences
- Awarded the best volunteer award for long-term contributions to the group

COURSES

- R Programming – R language for Absolute Beginners 2024
- R Programming – Advanced Analytics in R for Data Science 2024
- Data Science Bootcamp: Your First Step as a Data Scientist 2024
- Qualitative Research: Building on Foundations by Dr Audrey R. Giles

AWARDS & Scholarships

- Awarded Shield by Aga Khan Education Services for Result in F.Sc part I: 2017
- Awarded Shield by Aga Khan Education Services Pakistan for Results in FSc part II: 2018
- Runner-up for Best Dean's Clinical Research Fellow 2025
- Presidential Scholarship, Harvard T.H. Chan School of Public Health (full tuition coverage), 2026

TECHNICAL SKILLS

- Statistics on STATA: Proficient with descriptive analysis, Shapiro-Wilk tests, t-tests, Mann-Whitney U tests, chi-square tests, Pearson, Spearman's correlation tests, etc.
- Statistics on R: Proficient with descriptive and inferential analysis, Distributed Lagged Non-linear Models, Predictive Modeling, and Propensity Score Matching Analysis

- HTML and CSS: Skilled in the Basics of HTML and CSS, and website programming using both languages
- JavaScript: Know the basics of JavaScript and its application in website functioning
- Worked on improving the Quadragon website and adding new webpages.
- Adobe Photoshop: Skilled in Photo-editing and poster-making
- Adobe Premiere Pro: Skilled in Video Editing
- Other software: Skilled with EndNote, Google Sheets, MS Excel
- Climate and Health Data: Harnessing Satellite, Geospatial, and Ground Monitoring Data to Generate Evidence: 4-day workshop organized by the Department of Community Health Services, Aga Khan University
- Data Management: Managing national Emergency Medical Services data from Rescue 1122 Punjab, Pakistan. Experienced with data cleaning and sharing
- Grant Writing: Currently working on NIH R01 Grant Proposal under the mentorship of Dr. Junaid Razzak

LANGUAGES

Urdu (fluent); Hindi (fluent); Sindhi (fluent)